

DHTSystem

```
void addNode(NodeThread node)
nodes.add(node)
node.start()
void buildDiscoveryChain()
nodes.get(i).addPeer(nodes.get(i+1))
nodes.get(i+1).addPeer(nodes.get(i)); // Make it bidirectional
void printChain()
```

Main

```
putMsg.type = Message.Type.PUT
putMsg.key = "Colour"
putMsg.value = "Red"
putMsg.origin = n1
getMsg.type = Message.Type.GET
getMsg.key = "Colour"; //ok
getMsg.origin = n3

void main(String[] args) throws InterruptedException
system.addNode(n1)
system.addNode(n2)
system.addNode(n3)
system.buildDiscoveryChain()
system.printChain()
n1.receiveMessage(putMsg)
Thread.sleep(500)
n3.receiveMessage(getMsg)
```

Message

```
Type type
String key
String value
Node origin
int hops = 0
```

Node

```
void receiveMessage(Message msg)
long getId()
```

NodeThread

```
long id
boolean running = true
break
break
```

```
NodeThread(long id)
void run()
Message msg = mailbox.poll(1, TimeUnit.SECONDS)
processMessage(msg)
} catch (InterruptedException e)
Thread.currentThread().interrupt()
void processMessage(Message msg)
String originId = (msg.origin != null) ? String.valueOf(msg.origin.getId()) : "Unknown"
switch (msg.type)
localStore.put(msg.key, msg.value)
log("STORED key: " + msg.key + " (Origin: Node " + msg.origin.getId() + ")")
log("Key not found. Forwarding to peers...")
log("Passing request to Node " + peer.getId())
peer.receiveMessage(msg)
void addPeer(Node peer)
void receiveMessage(Message msg)
mailbox.offer(msg)
void log(String text)
```